Homework 1

In this homework, we learn how to use variational autoencoder (VAE) to generate medical images. First, you can use PyTorch to implement your own version of VAE. Then, take the chest X-ray dataset from `https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia` and train a VAE model on the chest X-ray dataset. Use the trained VAE to generate chest X-ray images, and write a report (2 pages, NeurIPS format). In the report, you can include the following: 1) hyperparameter settings; 2) training loss curves; 3) Inception score and FID score of generated images, using code from `https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/`
4) examples of generated images. Submit the report and code to Gradescope.